

Contact-Geometric Magnetic Levitation: The r^* Lock Gap — Whitney A_1 Fold as Stable Float Distance in a Rotating Disc Prototype

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Earnshaw's theorem (1842) proves that static fields cannot produce stable 3D equilibrium for a free rigid body. Circumventions exist — diamagnets, superconductors, rotating fields — but none identifies the geometric origin of the stable float distance. We address this gap using the contact 3-manifold $(\mathbb{R}^3, \alpha = dz - r^2 d\theta)$, where $\alpha = 0$ reads $dz = r^2 d\theta$: vertical lift is geometrically coupled to angular momentum. A disc satisfying $\dot{\theta} = 1$ obeys this constraint at every point. The governing ODE ($\varepsilon = 2$) is $\dot{r} = r(1 - r^2) + 2(r - 1)e^{-z}$, $\dot{\theta} = 1$, $\dot{z} = r^2 - 2(r - 1)^2 e^{-z}$. The unit helix $\Gamma = \{r = 1, \dot{\theta} = 1, \dot{z} = 1\}$ is a globally attracting limit cycle for $r(0) > r^*$ and $z(0) \geq \log 2$, convergence rate $\mu = -2$ (Theorems B.1–B.5, [1]).

The r^* lock gap. The inner basin boundary r^* is certified by adaptive bisection (DOP853, $\text{rtol} = 10^{-12}$): $r^* = 0.77594059$ (7 significant figures, reproducible on any IEEE-754 platform [2]). Stability hierarchy: $\varepsilon_0 = \frac{1}{3} < \frac{2}{3} < r^* = 0.77594 < \kappa^* \approx 0.882 < 1$. For $r < r^*$ trajectories are repelled; for $r > r^*$ they converge to Γ . The disc is therefore geometrically locked at r^* : the inner repulsive channel prevents collapse onto the lower field source; the outer attractive channel prevents escape. This is the contact-geometric resolution of Earnshaw.

Whitney A_1 singularity and Theorem B.3. We identify r^* as a Whitney A_1 fold singularity of the asymptotic radial map $\Phi: \mathbb{R}_+ \rightarrow \{1\}$. The Jacobian at the saddle (r^*, z^*) satisfies (Theorem B.3, proved in closed form in [1]):

$$\text{tr}(J)|_{\text{saddle}} = 2 \cos\left(\frac{2\pi}{7}\right) \approx 1.247, \quad \det(J) = -\varepsilon_0 = -\frac{1}{3}.$$

The negative determinant is the algebraic certificate of the bifurcation between the attractive and repulsive channels at r^* . The fold is topologically stable: small perturbations do not destroy it. In a rotating electromagnetic prototype — opposing magnetic poles above and below the disc — the fold boundary corresponds to the stable float gap, a structural consequence of the contact geometry rather than a fitted parameter.

Matter/dual-field duality. The system carries an exact $\varepsilon \rightarrow -\varepsilon$ duality: the outer basin switches from attractive ($\varepsilon = +2$, matter channel) to repulsive ($\varepsilon = -2$, dual channel). In a magnetic atom trap (Chu, Cohen-Tannoudji, Phillips, Nobel 1997) atoms lock at a stable radius between an attractive outer field and a repulsive inner field — the same geometry, the same fold. The ALPHA experiment (2023, [3]) confirms antihydrogen falls under gravity at $\approx 30\%$ precision; the repulsive channel at fine scales is not excluded. The contact geometry provides a framework for the threshold.

Open problem. A closed-form analytic expression for $r^* = 0.77594059$ does not yet exist. Lean 4 mechanisation is in progress (AXLE, github.com/TOTOGT/AXLE).

Keywords: contact geometry · magnetic levitation · Whitney A_1 singularity · Earnshaw's theorem · rotating electromagnetic system · basin of attraction

[1] P.N. Grossi, *Contact-Geometric Theory of Generative Transitions: Seven Proofs of the Tribonacci Constant*, doi:10.5281/zenodo.20682934 (2026).

[2] P.N. Grossi, *certify_rstar.py*, github.com/TOTOGT/3M (2026).

[3] ALPHA Collaboration, *Nature* 615, 591–595 (2023).